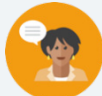


3.6 Evaluating Numerical Expressions with Integer Exponents

Lesson Introduction

 Benchmark	MA.8.NSO.1.3 Extend previous understanding of the laws of exponents to include integer exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational-number bases, with procedural fluency.		 Learning Target	I can evaluate a numerical expression using the laws of exponents.
 Benchmark Coherence	Building On MA.7.NSO.1.1 Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational bases.		Working Toward MA.912.NSO.1.1 Extend previous understanding of the laws of exponents to include rational exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.	
 Essential Questions	- How can you use the laws of exponents to evaluate numerical expressions? - Does order of operations apply when evaluating numerical expressions with exponents?			
 Lesson Narrative	Students continue building mathematical fluency in applying the laws of exponents to evaluate numerical expressions with exponents. It is reinforced for students that laws can be applied differently but generate equivalent expressions that have the same value. Students discover order of operations is still applicable when evaluating expressions with exponents.			
 Component Summary	3.6.1 Warm-Up (~5 min.)	Write equivalent expressions using only digits zero through nine (<i>SE p. 130</i>)	3.6.4 Guided Practice (10 min.)	Analyze and synthesize two students' processes used to evaluate expressions (<i>SE p. 132</i>)
	3.6.2 Guided Instruction (10-15 min.)	Explore different methods of evaluating exponential expressions (<i>SE p. 130</i>)	3.6.5 Collaboration (10-15 min.)	Evaluate numerical expressions (<i>SE p. 133</i>)
	3.6.3 Try It (5-10 min.)	Practice evaluating exponential expressions (<i>SE p. 131</i>)	3.6.6 Wrap-Up (~5 min.)	Identify and correct errors evaluating numerical expressions with rational bases and integer exponents (<i>SE p. 134</i>)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Zero exponent - Power of products - Identity exponent - Product of powers - Power of a power - Power of quotients - Negative exponent - Quotient of powers		(Optional) Highlighters (Optional) Chart paper (Optional) Markers	N/A

 Teacher Tips	Common Misconceptions - Students may not recognize that the order of operations still applies when simplifying expressions with exponents. - Students may believe there is a uniform order that should always be used to evaluate expressions instead of using more efficient strategies. In 3.6.2 Guided Instruction, students reflect on two different strategies to be able to recognize that while the strategies are different, both result in the same value.	Things to Consider - This lesson does not require calculators. - Consider exploring all correct ways the laws of exponents can be used to evaluate each expression. - Consider allowing students to use highlighters or markers and chart paper to complete 3.6.5 Collaboration.															
	Literacy and Language Routines Number Talk Consider implementing this routine during 3.6.3 Try It. After showing the problem, give students a few minutes to quietly think and give a signal when they have an answer and a strategy. Select students to share different strategies. Ask, “Who thought about it a different way?” Record explanations for all to see. Prompt students to provide more details about why they decided to approach a problem the way they did.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.5.1: Patterns and Structures In 3.6.5 Collaboration, students evaluate an expression using a different strategy than their partner. Then they discuss similarities and differences in their work. In doing so, students will recognize the patterns and structure in applying laws of exponents to evaluate numerical expressions.															
	 Differentiate Instruction	3.6.4 Guided Practice provides a visual entry point by having students highlight, circle, or underline each step the sample students took as they simplified the expressions. Provide an auditory entry point by reading the completed paragraph in question 1b aloud. Additional differentiation opportunities are denoted throughout the lesson guide.															
<table><tr><th>Danny’s Work</th><th>Reina’s Work</th></tr><tr><td>$\frac{\left(\frac{3}{4}\right)^{-3} \cdot \left(\frac{3}{4}\right)^0}{\left(\frac{3}{4}\right)^{-2}}$</td><td>$\frac{\left(\frac{3}{4}\right)^{-3} \cdot \left(\frac{3}{4}\right)^0}{\left(\frac{3}{4}\right)^{-2}}$</td></tr><tr><td>$= \frac{\left(\frac{3}{4}\right)^{-3+0}}{\left(\frac{3}{4}\right)^{-2}}$</td><td>$= \frac{\left(\frac{3}{4}\right)^{-3} \cdot 1}{\left(\frac{3}{4}\right)^{-2}}$</td></tr><tr><td>$= \frac{\left(\frac{3}{4}\right)^{-3}}{\left(\frac{3}{4}\right)^{-2}}$</td><td>$= \frac{\left(\frac{3}{4}\right)^{-3}}{\left(\frac{3}{4}\right)^{-2}}$</td></tr><tr><td>$= \left(\frac{3}{4}\right)^{-3-(-2)}$</td><td>$= \left(\frac{3}{4}\right)^{-3-(-2)}$</td></tr><tr><td>$= \left(\frac{3}{4}\right)^{-1}$</td><td>$= \left(\frac{3}{4}\right)^{-1}$</td></tr><tr><td>$= \frac{4}{3}$</td><td>$= \frac{4}{3}$</td></tr></table>			Danny’s Work	Reina’s Work	$\frac{\left(\frac{3}{4}\right)^{-3} \cdot \left(\frac{3}{4}\right)^0}{\left(\frac{3}{4}\right)^{-2}}$	$\frac{\left(\frac{3}{4}\right)^{-3} \cdot \left(\frac{3}{4}\right)^0}{\left(\frac{3}{4}\right)^{-2}}$	$= \frac{\left(\frac{3}{4}\right)^{-3+0}}{\left(\frac{3}{4}\right)^{-2}}$	$= \frac{\left(\frac{3}{4}\right)^{-3} \cdot 1}{\left(\frac{3}{4}\right)^{-2}}$	$= \frac{\left(\frac{3}{4}\right)^{-3}}{\left(\frac{3}{4}\right)^{-2}}$	$= \frac{\left(\frac{3}{4}\right)^{-3}}{\left(\frac{3}{4}\right)^{-2}}$	$= \left(\frac{3}{4}\right)^{-3-(-2)}$	$= \left(\frac{3}{4}\right)^{-3-(-2)}$	$= \left(\frac{3}{4}\right)^{-1}$	$= \left(\frac{3}{4}\right)^{-1}$	$= \frac{4}{3}$	$= \frac{4}{3}$	
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$= \frac{4}{3}$	$= \frac{4}{3}$																
Lesson Notes:																	

3.6.1 Warm-Up: Negative Exponents

Component Overview: Students use digits zero through nine to write a true numerical statement where one of the expressions in the statement has a negative exponent and the other expression is a fraction.



3.6.1 Warm-Up: Negative Exponents

1. Place a digit in each blue box to make the statement true. Use only digits 0 to 9, at most one time each.

$$\boxed{} - \boxed{} = \frac{\boxed{}}{\boxed{}\boxed{}}$$

Answers will vary.

$$7^{-2} = \frac{1}{49}$$



If time permits, have students share with a partner or whole group the expressions they wrote and the strategies they used. Be on the lookout for unique expressions if time is limited for sharing whole group.

3.6.2 Guided Instruction: Using the Laws of Exponents to Evaluate Expressions

Component Overview: Students analyze two students' work and describe the similarities and differences. Students determine if it is helpful to apply the laws of exponents before evaluating an expression.



3.6.2 Guided Instruction: Using the Laws of Exponents to Evaluate Expressions

- Melanie and Adonis each evaluated the expression $\frac{(5)^{-2}(5)^5}{5}$. Each student's steps are shown.

Melanie	Adonis
$\frac{(5)^{-2}(5)^5}{5}$	$\frac{(5)^{-2}(5)^5}{5}$
$= \frac{5^3}{5}$	$= \frac{\frac{1}{5^2} \cdot (5)^5}{5}$
$= 5^2$	$= \frac{\frac{1}{25} \cdot 3125}{5}$
$= 25$	$= \frac{125}{5}$
	$= 25$

Discuss with your partner how Melanie's steps and Adonis' steps are similar and different from each other. Then, summarize your discussion in the table.

Similarities	Differences
<i>Both students found the value to be 25. Both started by applying a law of exponent.</i>	<i>Melanie applied product of powers first and Adonis rewrote the negative exponent first. Melanie applied exponent laws until she had the fewest number of bases, and then evaluated. Adonis evaluated the exponents and then performed the operations.</i>

Answers will vary.



Consider prompting students to discuss whether they have a preferred strategy when comparing the two. Many students might point to Melanie's strategy as preferred because there are fewer steps, but that doesn't mean that with every expression, product of powers should be applied before negative exponents. In this case, though, it was more efficient. For other expressions, it may not be. The point is that both strategies are correct ones, and both resulted in the same value when laws were applied correctly.

3.6.2 Guided Instruction: Using the Laws of Exponents to Evaluate Expressions (continued)



When evaluating exponential expressions, it is helpful to apply the laws of exponents before evaluating.

2. Complete the steps to evaluate the expression $\frac{(6^2)^{-3}}{6^{-5}}$ by completing the description of each step and writing the resulting equivalent expression on the line that follows.

a. Apply
☐ product of power law.
☒ power of a power law.
 $\frac{6^{-6}}{6^{-5}}$

b. Apply
☒ quotient of power law.
☐ power of a power law.
 6^{-1}

c. Evaluate. $\frac{1}{6} \text{ or } 0.1\bar{6}$

3. Consider the expression $\left(\frac{4}{5}\right)^{-2} \cdot (4)^2$.

a. Discuss with your partner which laws of exponents could be used to evaluate the expression.

b. Evaluate $\left(\frac{4}{5}\right)^{-2} \cdot (4)^2$.

25



Facilitate a discussion focused on the impact of applying the laws of exponents and the order in which it's done for various expressions. It will be beneficial to display an anchor chart with the laws of exponents for students to reference.



For question 3, teachers can choose to guide students or to have students work in partners to evaluate. If the latter, there is an opportunity to share different students' processes and discuss/compare. For example, some students may take more steps depending on the order in which they apply the laws.

Focus of instruction should not be on naming the laws but rather the impact of applying them and the order in which it's done for various expressions. At this point, students will have worked with exponent laws in the 6th Grade Accelerated course and throughout this unit. There should be an anchor chart they can refer to in the room (or to the 3.1.2 Refresher in Lesson 1 in this unit if not).

3.6.3 Try It: Evaluating Expressions

Component Overview: Students try evaluating expressions independently and check off the exponent laws they use in the process of evaluating.



3.6.3 Try It: Evaluating Expressions

- Evaluate each expression. Show your work in the space provided. Then, check off the exponent laws applied to each expression from the list provided.

a. $\left(\frac{3}{4}\right)^{-2} \cdot 3$

$$\left(\frac{4}{3}\right)^2 \cdot 3$$

$$\left(\frac{4^2}{3^2}\right) \cdot 3$$

$$\left(\frac{4^2}{3}\right)$$

$$\frac{16}{3}$$

Exponent Laws Applied

- ☐ identity exponent
- ☒ negative exponent
- ☐ power of a power
- ☐ power of a product
- ☒ power of a quotient
- ☐ product of powers
- ☒ quotient of powers
- ☐ zero exponent



What order of steps did you take? Why did you choose this order of steps? What is another order you could have used? Try using that order and verify if the values are the same.

b. $\frac{(-7)^5}{(-7)^0(-7)^2}$

$$\frac{(-7)^5}{(-7)^2}$$

$$(-7)^3$$

$$-343$$

Exponent Laws Applied

- ☐ identity exponent
- ☐ negative exponent
- ☐ power of a power
- ☐ power of a product
- ☐ power of a quotient
- ☐ product of powers
- ☒ quotient of powers
- ☒ zero exponent



Students may benefit from using a highlighter or colored pencil to keep track of the changes they make at each step as they apply the laws of exponents.

3.6.4 Guided Practice: Evaluating Exponential Expressions

Component Overview: Students discuss and compare how two students' processes are used in evaluating the same expression. Students synthesize the processes used by each student by completing descriptions of each student's work.



3.6.4 Guided Practice: Evaluating Exponential Expressions

1. Danny and Reina evaluated the expression $\frac{\left(\frac{3}{4}\right)^{-3} \cdot \left(\frac{3}{4}\right)^0}{\left(\frac{3}{4}\right)^{-2}}$.

They found the same value but used different processes. Their work is shown.

Danny's Work	Reina's Work
$\frac{\left(\frac{3}{4}\right)^{-3} \cdot \left(\frac{3}{4}\right)^0}{\left(\frac{3}{4}\right)^{-2}}$ $= \frac{\left(\frac{3}{4}\right)^{-3+0}}{\left(\frac{3}{4}\right)^{-2}}$ $= \frac{\left(\frac{3}{4}\right)^{-3}}{\left(\frac{3}{4}\right)^{-2}}$ $= \left(\frac{3}{4}\right)^{-3-(-2)}$ $= \left(\frac{3}{4}\right)^{-1}$ $= \frac{4}{3}$	$\frac{\left(\frac{3}{4}\right)^{-3} \cdot \left(\frac{3}{4}\right)^0}{\left(\frac{3}{4}\right)^{-2}}$ $= \frac{\left(\frac{3}{4}\right)^{-3} \cdot 1}{\left(\frac{3}{4}\right)^{-2}}$ $= \frac{\left(\frac{3}{4}\right)^{-3}}{\left(\frac{3}{4}\right)^{-2}}$ $= \left(\frac{3}{4}\right)^{-3-(-2)}$ $= \left(\frac{3}{4}\right)^{-1}$ $= \frac{4}{3}$

- a. With your partner, discuss and compare how each student evaluated the expression.

Answers will vary. Discussions may include the difference in properties applied in the first step and the similarities in steps after that.



Use observational data from prior lessons to dictate how "guided" this 3.6.4 Guided Practice should be for you and your classroom. This could mean small-group instruction for some while others attempt independently, or it could mean teacher facilitation of practice with students actively working as a whole group.



Provide a visual entry point by having students highlight, circle, or underline each step the sample students took as they simplified the expressions. Provide an auditory entry point by reading the completed paragraph in Question 1b aloud.

3.6.4 Guided Practice: Evaluating Exponential Expressions (continued)

- b. Use words or phrases from the bank to complete the descriptions of each student's work. Some words or phrases will be used more than once and others not at all

zero	identity
power of a power	negative
power of a product	product of powers
power of a quotient	quotient of powers

Danny applied the product of powers exponent law first by rewriting $\left(\frac{3}{4}\right)^{-3} \cdot \left(\frac{3}{4}\right)^0$ as $\left(\frac{3}{4}\right)^{-3}$. Then, he applied the quotient of powers exponent law, subtracting the exponents -3 and -2 . Finally, he applied the negative exponent law to find the value of the expression.

Reina applied the zero exponent law first by re-writing $\left(\frac{3}{4}\right)^0$ as 1. Then, she multiplied $\left(\frac{3}{4}\right)^{-3}$ by 1. Next, she applied the quotient of powers exponent law by subtracting the exponents -3 and -2 . Finally, she applied the negative exponent law to find the value of the expression.

While the students applied the laws differently, they both generated equivalent expressions in each step to find the value of the expression.

3.6.5 Collaboration: Choose Your Own Adventure!

Component Overview: Students collaboratively determine different strategies that can be used to evaluate expressions. They discuss similarities and differences in the different strategies they used to evaluate the expressions. Students find that when the laws of exponents are applied correctly, the resulting value will be the same.



3.6.5 Collaboration: Choose Your Own Adventure!

1. Consider the expression $\frac{(2^{-1})^3 \cdot (\frac{1}{2})^{-4}}{2 \cdot 2^2}$.
- a. Discuss with your partner which exponent laws could be applied in the first step. Check off all that apply from the list.

- | | |
|--|--|
| ☐ Zero Exponent | ☐ Product of Powers |
| ☐ Identity Exponent | ☐ Power of a Product |
| ☐ Negative Exponent | ☐ Quotient of Powers |
| ☐ Power of a Power | ☐ Power of a Quotient |

- b. Evaluate the expression beginning with a different first step than your partner.

Answers will vary.

$$\frac{(2^{-1})^3 \cdot (\frac{1}{2})^{-4}}{2 \cdot 2^2}$$

$$\frac{(2^{-1})^3 \cdot (2)^4}{2 \cdot 2^2}$$

$$\frac{2^{-3} \cdot 2^4}{2 \cdot 2^2}$$

$$\frac{2}{2 \cdot 2^2}$$

$$\frac{2}{2 \cdot 4}$$

$$\frac{2}{8}$$

$$\frac{1}{4}$$

- c. Compare your work with your partner's and come to a consensus about the value of the expression. Discuss the similarities and differences in your work.



Consider displaying an anchor chart with examples of each law of exponents for students to reference.



Monitor while students are working to determine if there are any questions that need to be debriefed before moving to the next component. This could include highlighting student voices to share rich discussions or comments overheard while they collaborated for the whole class to hear or to address common misconceptions that were evident among the majority of the class.



Encourage students to explain why each law could or could not be applied first. Guide students to make the connection to recognize that they must use the order of operations when simplifying expressions.

3.6.5 Collaboration: Choose Your Own Adventure! (continued)

2. Consider the expression $\frac{\left(\frac{3}{2}\right)^{-5} \cdot \left(\left(\frac{2}{3}\right)^7\right)^0}{\left(\left(\frac{2}{3}\right)^2\right)^3}$.

- a. Discuss with your partner which exponent laws could be applied in the first step. Check off all that apply from the list.

- | | |
|--|--|
| ☐ Zero Exponent | ☐ Product of Powers |
| ☐ Identity Exponent | ☐ Power of a Product |
| ☐ Negative Exponent | ☐ Quotient of Powers |
| ☐ Power of a Power | ☐ Power of a Quotient |

- b. Evaluate the expression beginning with a different first step than your partner.

Answers will vary.

$$\frac{\left(\frac{3}{2}\right)^{-5} \cdot \left(\left(\frac{2}{3}\right)^7\right)^0}{\left(\left(\frac{2}{3}\right)^2\right)^3}$$

$$\frac{\left(\frac{3}{2}\right)^{-5} \cdot 1}{\left(\left(\frac{2}{3}\right)^2\right)^3}$$

$$\frac{\left(\frac{3}{2}\right)^{-5}}{\left(\left(\frac{2}{3}\right)^2\right)^3}$$

$$\frac{\left(\frac{3}{2}\right)^5}{\left(\frac{2}{3}\right)^6}$$

$$\left(\frac{2}{3}\right)^{-1}$$

$$\frac{3}{2}$$

- c. Compare your work with your partner's and come to a consensus about the value of the expression. Discuss the similarities and differences in your work.



Students may benefit from sentence stems such as:

- "Our methods were the similar because we both..."
- Our methods were different, because...

3.6.6 Wrap-Up: Error Analysis

Component Overview: Students identify and correct errors in two students' work.



3.6.6 Wrap-Up: Error Analysis

- Two students each evaluated an expression. Their work is shown in the table. Each student made a mistake. Circle the step where each student made an error and describe the mistake they made.

Problem	Describe the Mistake
$\left(\frac{1}{5}\right)^4 \cdot \left(\frac{1}{5}\right)^{-2} \cdot 5^{-1}$ $= \left(\frac{1}{5}\right)^{-8} \cdot 5^{-1}$ $= 5^8 \cdot 5^{-1}$ $= 5^7$ $= 78,125$	The student multiplied the exponents instead of adding them.
$\left(\frac{3}{2}\right)^{-3} \cdot (2)^3$ $= \frac{2^{-3}}{3^{-3}} \cdot 2^3$ $= \frac{2^0}{3^{-3}}$ $= \frac{1}{3^{-3}}$ $= 3^3$ $= 27$	The student wrote the reciprocal of $\frac{3}{2}$ but did not use the opposite exponent.



The information provided from the students here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.



Prompt students who finish early to write a letter providing advice to each student to help them avoid making the same mistake again in the future.